



Graphing special number sequences using technology

Task A: Plotting sequences using Desmos

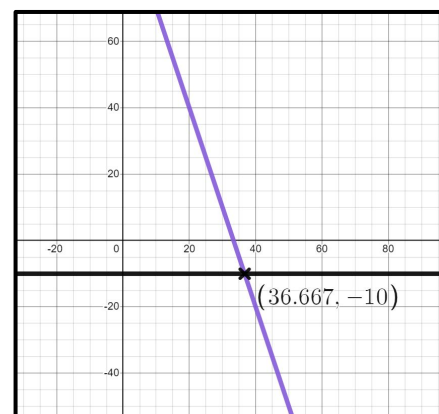
- 1) Lucas wants to know if -10 is in the sequence $100 - 3n$



I have plotted the lines $y = 100 - 3n$ and $y = -10$. They intersect so -10 is a term in the sequence $100 - 3n$

- a) What has Lucas done correctly?

He has plotted the correct lines and looked for the intersection point.



Lucas

- b) Explain to Lucas what mistake he has made.

To be in the sequence, the x coordinate has to be an integer. This coordinate shows -10 is between the 36^{th} and 37^{th} term but is not in the sequence.

- 2) Use Desmos to match these sequences to their n^{th} term rules

- a) Geometric sequence with first five terms:

2, 4, 8, 16, 32, ...

- b) Geometric sequence with first five terms:

8, 16, 32, 64, 128, ...

- c) Geometric sequence with first five terms:

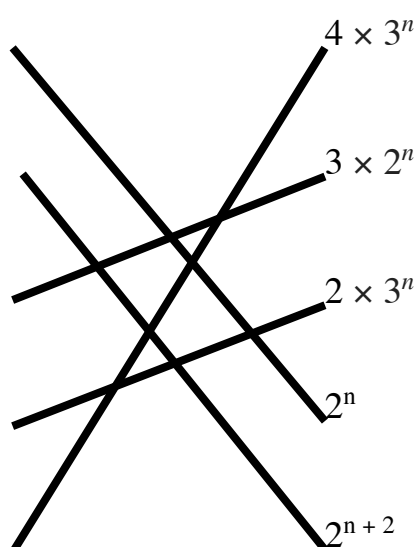
6, 12, 24, 48, 96, ...

- d) Geometric sequence with first five terms:

6, 18, 54, 162, 486, ...

- e) Geometric sequence with first five terms:

12, 36, 108, 324, 972, ...



3) Use graphing software to help with the following questions.

a) Is the number 30 in this sequence?

Yes the 16th term

Term number	1	2	3	4	n
Term	-1.5	0.6	2.7	4.8	$2.1n - 3.6$

b) Is the number 2044 in this sequence?

No between the 6th and 7th term

Term number	1	2	3	4	n
Term	8	24	72	216	$8 \times 3^{n-1}$

c) Is the number 1280 in this sequence?

Yes the 8th term

Term number	1	2	3	4	n
Term	10	20	40	80	5×2^n

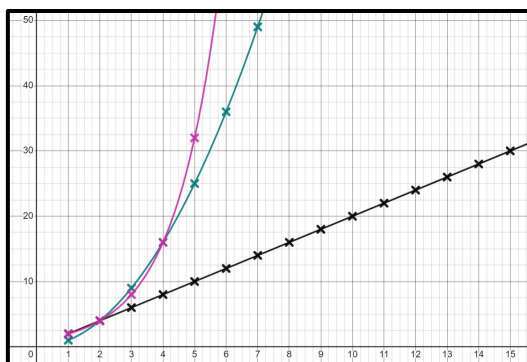
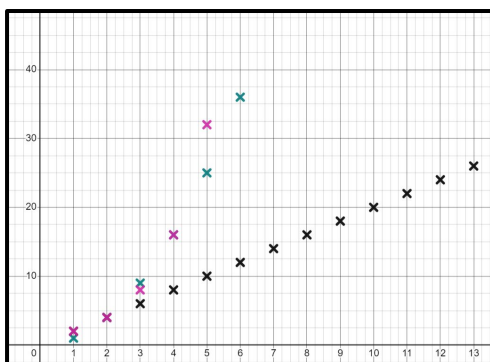
d) Is the number 4004 in this sequence?

No between the 5th and 6th term

Term number	1	2	3	4	n
Term	4	16	64	256	4^n

Task B: Exploring special number sequences

1a) Plot the first 20 points of the sequences $2n$, n^2 and 2^n



Adding the curves in may help see key features of these sequences

b) Two of the sequences have the same first term, which two? $2n$ and 2^n

c) What is happening at the coordinate (2,4) what does this mean about the sequences?

All three sequences have a second term of 4

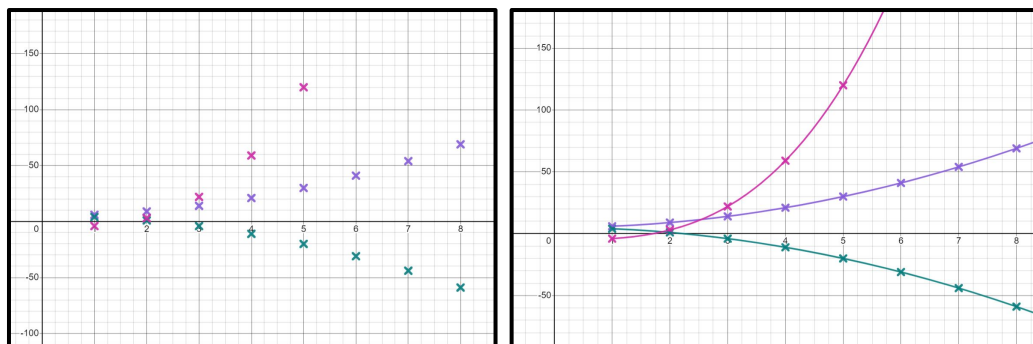
d) Which sequence is increasing at the fastest rate? 2^n

e) Find four numbers which are in the sequence $2n$ and 2^n

all numbers in 2^n are also in $2n$ so 2, 4, 8, 16, 32, 64, 128,.....

f) What is the 15th term in the sequence 2^n ? 32 768

2a) Plot the first 20 points of the sequences $5 - n^2$, $5 + n^2$ and $n^3 - 5$



Adding the curves in may help see key features of these sequences

b) Which sequence has the smallest 1st term? $n^3 - 5$

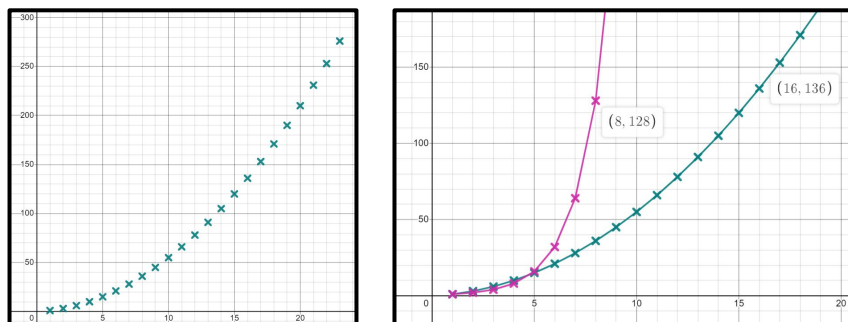
c) Which sequence has the smallest 2nd term? $5 - n^2$

d) Which sequence has the largest 3rd term? $n^3 - 5$

e) What do you notice about the sequences $5 + n^2$ and $5 - n^2$?

You may have said: they are reflections of each other, the terms in $5 + n^2$ are always larger than the terms in $5 - n^2$

3a) The sequence of all triangular numbers has the n^{th} term rule $\frac{n(n+1)}{2}$ plot the first 20 terms.



b) What is the 19th triangular number?

190

c) The geometric sequence which starts 1, 2, 4, 8,... has the n^{th} term rule 2^{n-1} at what term number does this sequence overtake the triangular numbers?

Term 5

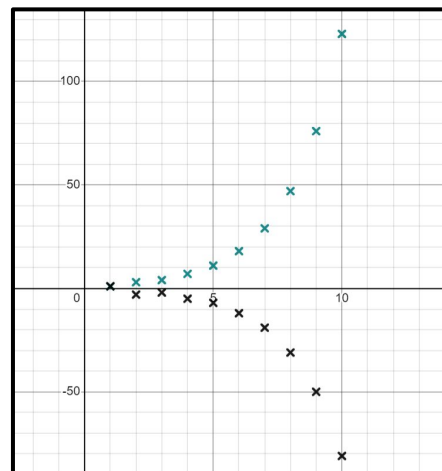
e) Which is larger the 8th term in the sequence 2^{n-1} or the 16th triangular number.

The 16th triangular number ($136 > 128$)

4a) A sequence starts 1, 3, 4, ... the next term is found by adding the two previous terms together. Plot the first 10 terms of this sequence.

b) A different sequence starts 1, -3, ... the next term is found by adding the two previous terms together. Plot the first 10 terms of this sequence.

c) How does the shape of the graph of this sequence differ to the previous one?



E.g. After the 3rd term the graph is decreasing. After the first few terms both graphs become curves, the second graph has a general curve shape like the previous one but with a noticeable 'dip' at term 2

